Fall 1990

Problem 1 (Fa90). Find all pairs of integers a and b satisfying $0 < a < b$ and $a^b = b^a$.

Problem 2 (Fa90). Evaluate the integral $\dfrac{1}{2\pi i}\displaystyle\int_C \frac{\mathrm{d}z}{(z-2)(1+2z)^2(1-3z)^3}$ where C is the circle $|z| = 1$ with counterclockwise orientation.

Problem 3 (Fa90). Let R be a ring with identity, and let $\mathfrak{I}$ be the left ideal of R generated by $\{ab - ba \mid a, b \in R\}$. Prove that $\mathfrak{I}$ is a two-sided ideal.

Problem 4 (Fa90). Suppose f is a continuous real valued function. Show that

$$\int_0^1 f(x)x^2\,\mathrm{d}x = \frac{1}{3}f(\xi)$$

for some $\xi \in [0, 1]$.

Problem 5 (Fa90). Let A be a real symmetric $n \times n$ matrix that is positive definite. Let $y \in \mathbb{R}^n$, $y \neq 0$. Prove that the limit

$$\lim_{m\to\infty} \frac{y^\top A^{m+1} y}{y^\top A^m y}$$

exists and is an eigenvalue of A.

Problem 6 (Fa90, Fa22). Let the function f be analytic in the entire complex plane, and suppose that $f(z)/z \to 0$ as $|z| \to \infty$. Prove that f is constant.

Problem 7 (Fa90). Let G be a group and N be a normal subgroup of G with $N \neq G$. Suppose that there does not exist a subgroup H of G satisfying $N \subset H \subset G$ and $N \neq H \neq G$. Prove that the index of N in G is finite and equal to a prime number.

Problem 8 (Fa90, Sp97). Let f be a continuous real valued function satisfying $f(x) \geqslant 0$, for all x, and

$$\int_0^\infty f(x)\,\mathrm{d}x < \infty$$

Prove that

$$\frac{1}{n}\int_0^n xf(x)\,\mathrm{d}x \to 0$$

as $n \to \infty$.

Problem 9 (Fa90). Let $\mathbb{R}^3$ be 3-space with the usual inner product, and $(a, b, c) \in \mathbb{R}^3$ a vector of length 1. Let W be the plane defined by $ax + by + cz = 0$. Find, in the standard basis, the matrix representing the orthogonal projection of $\mathbb{R}^3$ onto W.

Problem 10 (Sp78, Sp82, Su82, Fa90). Determine the Jordan Canonical Form of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 4 \end{pmatrix}$$

Problem 11 (Fa90). Suppose that f maps the compact interval I into itself and that

$$|f(x) - f(y)| < |x - y|$$

for all $x, y \in I, x \neq y$. Can one conclude that there is some constant $M < 1$ such that, for all $x, y \in I$,

$$|f(x) - f(y)| \leqslant M|x - y|?$$

Problem 12 (Fa90). Let A be an additively written abelian group, and $u, v : A \to A$ homomorphisms. Define the group homomorphisms $f, g : A \to A$ by

$$f(a) = a - v(u(a)) \qquad g(a) = a - u(v(a)) \qquad \text{with} \quad a \in A$$

Prove that the kernel of f is isomorphic to the kernel of g.

Problem 13 (Fa79, Fa90). Suppose that f is analytic on the open upper half-plane and satisfies $|f(z)| \leqslant 1$ for all z, $f(i) = 0$. How large can $|f(2i)|$ be under these conditions?

Problem 14 (Fa90, Fa22). Prove that $\sqrt{2} + \sqrt[3]{3}$ is irrational.

Problem 15 (Fa90). Let n be a positive integer and let P_{2n+1} be the vector space of real polynomials whose degrees are, at most, $2n + 1$. Prove that there exist unique real numbers $c_1, \ldots, c_n$ such that, for all $p \in P_{2n+1}$.

$$\int_{-1}^{1} p(x)\,\mathrm{d}x = 2p(0) + \sum_{k=1}^{n} c_k(p(k) + p(-k) - 2p(0))$$

Problem 16 (Fa90). Evaluate the limit $\lim_{n\to\infty} \cos\frac{\pi}{2^2}\cos\frac{\pi}{2^3}\cdots\cos\frac{\pi}{2^n}$.

Problem 17 (Fa90). Does the set $G = \{a \in \mathbb{R} \mid a > 0, a \neq 1\}$ form a group with the operation $a * b = a^{\log b}$?

Problem 18 (Fa90, Sp24). Let the function f be analytic in the entire complex plane and satisfy

$$\int_0^{2\pi} |f(re^{i\theta})|\, \mathrm{d}\theta \leqslant r^{17/3}$$

for all $r > 0$. Prove that f is the zero function.